

Lesson 0 — How to Learn: From Working Memory to Long-Term Memory

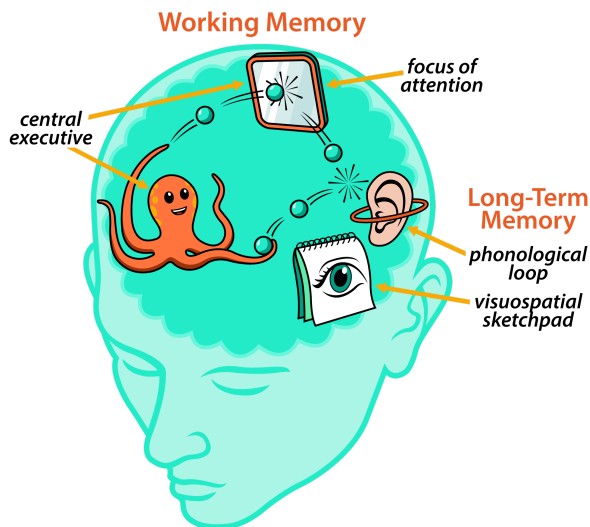
Name: _____ Date: _____

Purpose	Understand <i>why</i> we stop after every block to recall, and learn the routine you will use for the whole course
Key idea	Active recall + spacing + sleep turn what you understood today into what you still know on test day
When you use it	After every block in class · the same day after class · every night before sleep · a few days later
Source	Barbara Oakley, Beth Rogowsky & Terrence Sejnowski, <i>Uncommon Sense Teaching</i> (2021) — all figures

Why this matters in AP Calculus BC

BC has dozens of formulas: derivatives of trig and inverse trig functions, integration rules, series tests, Taylor series, polar and parametric formulas. You cannot look any of them up on the exam. And while you solve a hard problem, your brain has **very little room** for thinking. If every formula is stored in long-term memory, and you know **why** it is true, that room is free for the actual problem. This lesson shows you how to put formulas there so they stay.

1. Your Brain Has Two Memory Systems

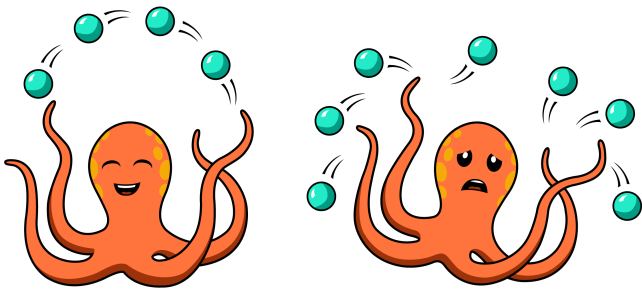


Working memory is what you are thinking about *right now*. Oakley pictures it as an **octopus** with only a few arms. It can hold about **four** pieces of information at a time, and only for a short while. If you stop paying attention, it lets go.

Long-term memory is your storage. It is huge and can keep things for years. But information only gets there if you **build** it there, and you can only **use** it if you can **find** it again.

Learning is moving knowledge from working memory into long-term memory, then practicing until you can pull it back out quickly.

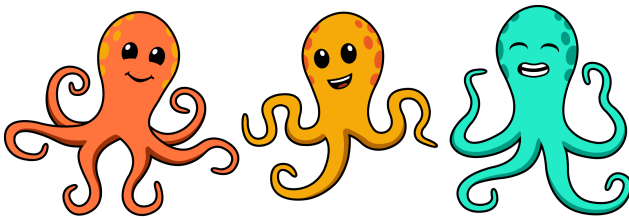
Too much at once → the octopus drops the balls



When a teacher gives too many new ideas in a row, working memory overloads (right). New ideas push out old ones, and you leave class remembering very little, even though every step "made sense."

That is why we stop after each block to recall. It empties the octopus's arms into long-term memory before the next block arrives.

Everyone's octopus is different



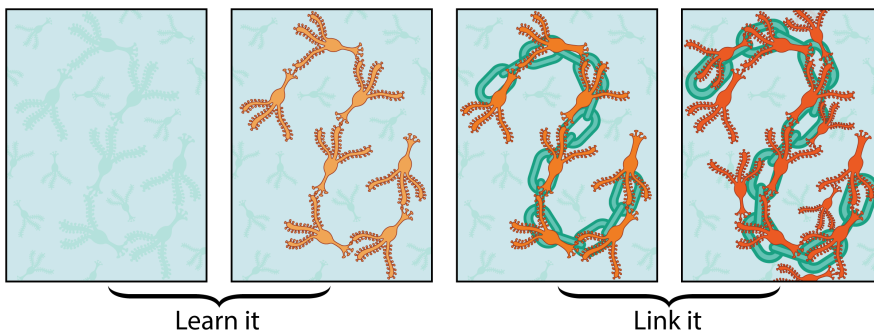
Some people's working memory holds more, some less. **This is normal and is not the same as being "smart" or "not smart."**

- If your working memory is **smaller**, you are forced to simplify and connect ideas. That often leads to deeper understanding, but you need to practice until the ideas are in long-term memory.
- If your working memory is **larger**, things feel easy in class, so it is tempting to skip practice. Then the ideas never reach long-term memory, and they are gone by test day.

Active recall works for both. Everyone does it.

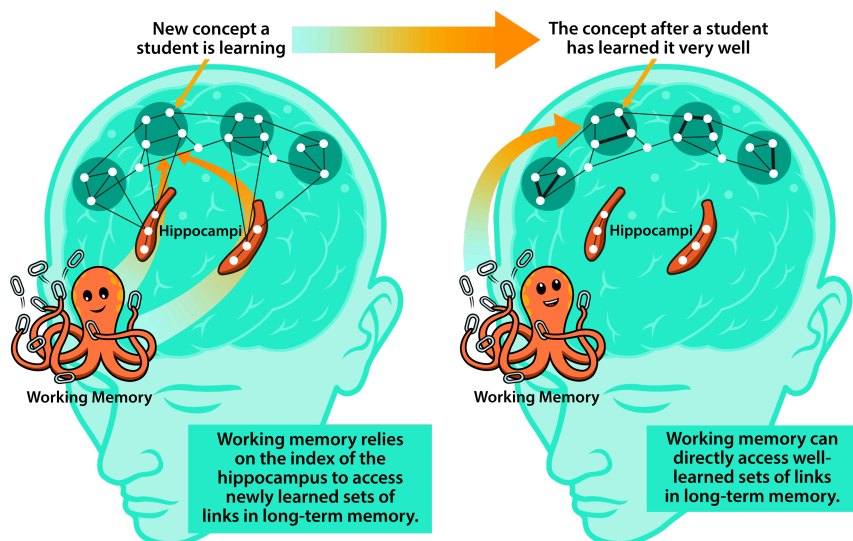
2. How a Long-Term Memory Is Built

Learn it, link it



A memory is a set of **linked neurons**. The first time you learn something, the links are weak (second panel). **Each time you pull the idea back out of your own head, the links get stronger and thicker** (third and fourth panels). Reading the idea again does much less; **retrieving** it is what builds the chain.

From "looking it up" to "just knowing"



- **Left:** When something is new, working memory has to go through the **hippocampus**, which acts like an index or table of contents, to find the new links. This is slow and fragile. It is like knowing that "the formula is somewhere in my notes."
- **Right:** After repeated recall over several days, working memory reaches the long-term links **directly**. You "just know" it, instantly, and it costs almost no working memory.

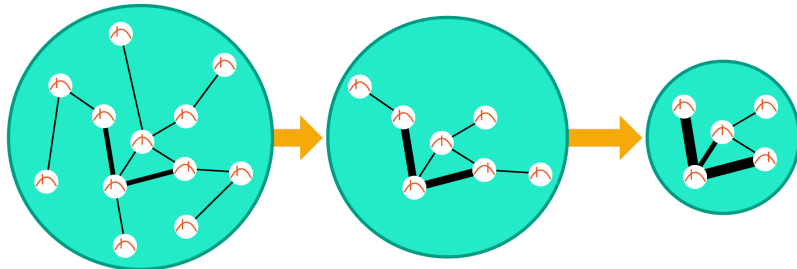
Our goal for every key formula: move it from the left picture to the right picture.

Messy at first, clean later

As students first learn a concept, their neural connections in long-term memory are disorderly and the arrangement isn't as simple as it could be.

Neurons in long-term memory consolidate over time as the hippocampus helps them rearrange into simpler, more efficient connections.

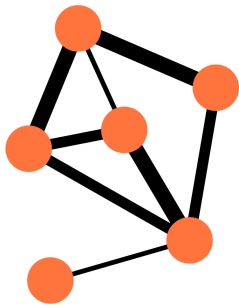
The consolidated long-term memory in the neocortex can eventually be accessed with no help from the hippocampus.



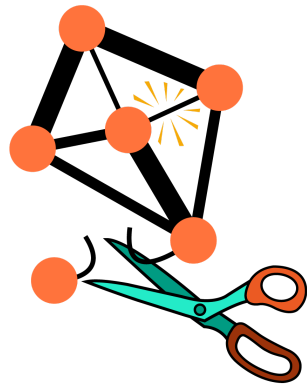
When you first learn a topic, the connections are messy. With time, recall, and sleep, the brain **consolidates** them into a smaller, simpler, more efficient network. That is why a topic that felt confusing on Monday often feels clearer on Thursday, **if you recalled it in between.**

3. Sleep Is Part of Studying

Before sleep



After sleep



While you sleep, your brain replays what you worked on that day. **Strong links get stronger, weak links get cut** (the scissors). New connections can even appear (the spark).

So:

- **Recall right before you go to sleep.** It tells your brain *"this is important, keep it."*
- **Don't give up sleep to study.** An all-nighter skips the step that saves what you learned.
- Studying a little every day, with sleep in between, beats studying a lot on one day.

4. Active Recall: What It Is and What It Is Not

Active recall means **pulling information out of your own memory, without looking**, and then checking.

❌ Feels like studying (but is weak)	✅ Active recall (strong)
Re-reading the handout	Close the handout; write everything you remember on a blank page
Highlighting	Explain the idea out loud, as if teaching a friend
Looking at a solution and thinking "yes, I get it"	Cover the solution; solve the example yourself; then compare
Copying formulas onto a sheet	Write the formula from memory and derive or explain why it is true
Watching a video again	Pause and predict the next step before it is shown

Illusion of competence. When the answer is in front of you, your brain says *"I know this."* That feeling is recognition, not recall. The test won't show you the answer. The only real check is: **can I produce it from a blank page?**

Recall should feel a little hard. **That struggle is the learning happening.** If you get stuck, that's fine: look, then close the page and try again.

5. The Routine for This Course (Do This Every Week)

① After each block in class — 2 to 3 minutes

1. **Close** the handout and notes.
2. On blank paper, write:
 - the key **definitions / formulas** of the block,

- **why** each one is true or where it comes from,
- **one small example** you make up yourself.

3. **Open** the handout and check. With a **different color pen**, fix and circle what you missed or got wrong.

② Same day, after class — 10 minutes

Before starting homework, recall the **whole lesson** on one page: the block titles, then the key ideas of each block. Check and circle gaps.

③ Before sleep — 5 minutes (the most important one)

No notes. Close your eyes or use a blank page and recall today's main ideas and formulas. Then sleep.

④ Spaced review — 5 to 10 minutes each

Recall the same lesson again **the next day**, **about 3 days later**, and **one week later**. The weekly warm-up quiz and Part D (spiral review) of every homework are built to help with this.

⑤ Keep your error log

Anything you missed twice goes into your **error log** (*topic · what I did wrong · the correct idea*). Recall the error log before every checkpoint and mock exam.

6. Recall With Reasoning, Not Just Memorizing

A formula stored **alone** is easy to forget and easy to mix up (a sign flips, a 2 goes in the wrong place). A formula stored **together with its reason** is linked to other things you know, so it is easier to find and you can **rebuild** it if it slips.

Example 1 (Lesson 1). Don't just memorize $y = a f(b(x - h)) + k$. Recall the reasoning:

- k is added to the **output**, so the graph moves **up** k .
- $x - h$ changes the **input**: to get the same output as before, x must be h **bigger**, so the graph moves **right** h . (That is why the sign "looks backwards.")
- a multiplies the output $\rightarrow$ vertical stretch by $|a|$; if $a < 0$, reflect over the x -axis.
- b multiplies the input $\rightarrow$ horizontal compression by a factor of $\frac{1}{|b|}$; if $b < 0$, reflect over the y -axis.

Example 2 (BC preview). Instead of memorizing $\frac{d}{dx} \tan x = \sec^2 x$ as a separate fact, remember **one** reason: $\tan x = \frac{\sin x}{\cos x}$. The quotient rule gives

$$\frac{d}{dx} \tan x = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

Example 3 (BC preview). $\cos^2 x = \frac{1 + \cos 2x}{2}$ is needed to integrate $\cos^2 x$. You don't have to memorize it separately: start from $\cos 2x = 2 \cos^2 x - 1$ and solve for $\cos^2 x$.

When you recall a formula, always ask yourself: **"Where does this come from? What would happen if I changed a sign?"** If you can answer, the formula is truly yours.

7. Recall Sheet (copy this format)

Block ____	
Key ideas / definitions (from memory)	
Formulas + why they are true	
My own example	
What I missed (different color, after checking)	

Summary

1. **Working memory is small** (about 4 items) and different for everyone. Too much new information at once → it overflows.
2. **Long-term memory is built by retrieval.** Every time you recall an idea from your own head, its links get stronger.
3. **With repeated recall over several days**, you stop needing to "look it up" in your head and just know it.
4. **Sleep saves what you practiced.** Recall before bed; never trade sleep for cramming.
5. **Routine:** recall after every block → recall the lesson the same day → **recall before sleep** → recall again after 1, 3, and 7 days.
6. **Recall with reasoning:** store every formula together with *why* it is true.

Figures: Barbara Oakley, Beth Rogowsky & Terrence J. Sejnowski, *Uncommon Sense Teaching: Practical Insights in Brain Science to Help Students Learn* (TarcherPerigee, 2021), Figures 3, 5, 7, 9, 12, 16, 17. Used for classroom instruction.